

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Compound Interest & Depreciation

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COMPOUND INTEREST & DEPRECIATION · CH1 · L7

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The Map

This lesson collapses Compound Interest & Depreciation into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 The multiplier for compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 32 website questions, 32 checked solutions, 3 local PDFs read.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| $0.85V$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q14

Let the original house value

This is a reasoning step: write one clear sentence, then continue the calculation.

Use compound interest

| $0.85V$

Finish: 10%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

14 Simon bought a house at the beginning of 2018 The value of Simon's house had decreased by 15% by the end of 2018 The house increased in value during both 2019 and 2020 The percentage increases in the value of the house during 2019 and 2020 were the same. The value of Simon's house at the end of 2020 was 2.85% greater than the amount he paid for his house at the beginning of 2018 Calculate the percentage increase in the value of the house during 2019 %

YOUR SOLUTION

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| $0.85V$

SIMPLEST STRATEGY

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WORKED MODEL

Source pattern: Jan 2021 P2HR Q14

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Finish: 10%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

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03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6000(1.015)^2 = 6181.35$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q11

Use compound interest

$$6000(1.015)^2 = 6181.35$$

Let the third year interest

$$6181.35 \left(1 + \frac{r}{100}\right) = 6311.16$$

Finish: 2.1%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

11 Max invests \$6000 in a savings account for 3 years. The account pays compound interest at a rate of 1.5% per year for the first 2 years. The compound interest rate changes for the third year. At the end of 3 years, there is a total of \$6311.16 in the account. Work out the compound interest rate for the third year. Give your answer correct to 1 decimal place. %

YOUR SOLUTION

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2020 \text{ value} = \frac{634400}{1.04} = 610000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q6

Use compound interest

$$2020 \text{ value} = \frac{634400}{1.04} = 610000$$

Use compound interest

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) No, it stayed in band B. (b) 27.75%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

6 Asha bought an apartment. The table gives information about the value of apartments, in euros, and the annual service charge band. Value (x euros) Service charge band
 $x < 700\,000$ A $600\,000 < x < 700\,000$ B $500\,000 < x < 600\,000$ C $400\,000 < x < 500\,000$ D $0 < x < 400\,000$ E
 In 2021, the value of Asha's apartment was 634 400 euros. The value of Asha's apartment had increased by 4% from its value in 2020 (a) Has the annual service charge band changed for Asha's apartment? Show your working clearly.
 (3) Pam bought a ...

YOUR SOLUTION

05 Use compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6000(1.015)^2 = 6181.35$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q11

Use compound interest

$$6000(1.015)^2 = 6181.35$$

Let the third year interest

$$6181.35 \left(1 + \frac{r}{100}\right) = 6311.16$$

Finish: 2.1%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 The multiplier for compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

1.025

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2HR Q5

The multiplier for compound interest

1.025

Use compound interest

$$7200(1.025)^3 = 7753.6125$$

Finish: 7754 dollars

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

5 Shane invests 7200 dollars for 3 years in a savings account. He gets 2.5% per year compound interest. How much money will Shane have in his savings account at the end of 3 years? Give your answer to the nearest dollar. dollars

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use percentage multiplier
"work out", "show", "hence", a familiar exam phrase	Reverse a percentage
"work out", "show", "hence", a familiar exam phrase	Use compound change
"work out", "show", "hence", a familiar exam phrase	Compare percentage change
"work out", "show", "hence", a familiar exam phrase	Use compound interest
"work out", "show", "hence", a familiar exam phrase	The multiplier for compound interest

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.